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BAD JOBS IN AMERICA: STANDARD AND NONSTANDARD EMPLOYMENT RELATIONS AND JOB QUALITY IN THE UNITED STATES

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The prevalence of nonstandard jobs is a matter of concern if, as many assume, such jobs are bad. We examine the relationship between nonstandard employment (on-call work and day labor, temporary-help agency employment, employment with contract companies, independent contracting, other self-employment, and part-time employment in "conventional" jobs) and exposure to "bad" job characteristics, using data from the 1995 Current Population Survey. Of workers age 18 and over, 31 percent are in some type of nonstandard employment. To assess the link between type of employment and bad jobs, we conceptualize "bad jobs" as those with low pay and without access to health insurance and pension benefits. About one in seven jobs in the United States is bad on these three dimensions. Nonstandard employment strongly increases workers' exposure to bad job characteristics, net of controls for workers' personal characteristics, family status, occupation, and industry.

A RECURRING theme in the sociology and economics of industrial societies is the problem of marginalized workers in bad jobs (Gordon 1972; Marx [1867] 1967; Morse 1969). In the late 1960s and early 1970s, institutional economists drew attention to the issue of bad jobs by proposing that the labor market was divided into a primary segment, comprising "good" jobs, and a secondary segment comprising "bad" jobs (Bluestone 1970; Doeringer and Piore 1971;

Harrison 1972; Kalleberg and Sørensen 1979; Piore 1971). Recent reports of the growth of "contingent" (Barker and Christensen 1998; Freedman 1985; Polivka and Nardone 1989), "externalized" (Pfeffer and Baron 1988), and nonstandard employment (Blank 1998; Casey 1991) again draw attention to bad jobs.

Underlying the concern with nonstandard work arrangements are claims that such jobs are growing rapidly and are worse for workers than regular full-time jobs (Appelbaum 1992; Callaghan and Hartmann 1991; but see Blank 1998). Whether the growth of nonstandard employment is problematic depends on the quality of nonstandard jobs. Some have suggested that nonstandard arrangements benefit both employers and workers (Belous 1989). The flexibility inherent in many types of nonstandard work allows employers to cut labor costs during slack times, and employers can screen nonstandard workers before hiring them permanently, thus re-

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ducing recruitment and training expenses (Houseman 1997). Workers also benefit insofar as nonstandard jobs let them control their schedules, sample a variety of jobs, and have more time for other activities (Belous 1989; Polivka and Nardone 1989).

To the extent that nonstandard jobs pay poorly, lack health insurance and pension benefits, are of uncertain duration, and lack the protections that unions and labor laws afford, they are problematic for workers (Ferber and Waldfogel 1996; Kalleberg et al. 1997; Mishel, Bernstein, and Schmitt 1999). Whether, on balance, nonstandard work primarily benefits or harms workers is a matter of both debate and considerable research. We advance this debate by directing the question away from the prevalence of nonstandard jobs—which have been taken as a proxy for marginal jobs—toward the link between these arrangements and the quality of jobs.

That bad jobs and nonstandard employment relations exist is nothing new. Marginal and irregular work and nonstandard employment relations were common among the laboring classes in industrial countries in the nineteenth century. Until the end of the Great Depression, most jobs were insecure and most wages were unstable (Jacoby 1985). Among the working classes, pensions and health insurance were almost unheard of before the 1930s, and benefits were contingent on workers' docility rather than entitlements (Edwards 1979). Laws enacted during the 1930s dramatically increased the number of workers whose jobs provided a living wage, employment security, and benefits (Dobbin et al. 1988). Workers' right to bargain collectively, along with increased government control over working conditions and employment practices, restricted employers' power over the terms of employment. Thus, by the 1940s, the ratio of good jobs to bad jobs had increased sharply (Edwards 1979; Jacoby 1985).

Although regular full-time jobs were the norm by the 1950s, nonstandard work arrangements and bad jobs had not disappeared. Employers continued to rely on a peripheral workforce to contain labor costs and as a buffer protecting the jobs of "permanent" employees (Harrison 1994; Morse 1969; Szymanski 1976). In fact, the growth of "good" jobs that are covered by collec-

tive bargaining and federal labor-protection laws made the "badness" of peripheral jobs even more striking. Still, the question of the link between employment structures and job quality remains.

To examine the relationship between nonstandard employment and "bad" jobs, we use data from the February 1995 Current Population Survey's (CPS) Supplement on Contingent Work (U.S. Census Bureau 1996).¹ The CPS is a monthly household survey of Americans. The 1995 supplement on contingent and alternative work arrangements provides the first systematic data on nonstandard employment in the United States.

STANDARD AND NONSTANDARD EMPLOYMENT RELATIONSHIPS

The term "nonstandard employment relationship" implies the existence of a "standard employment relationship," although there is no consensus on exactly what this standard relationship entails. Indeed, the meaning of the terms "employer" and "employee" are socially constructed in a political process (Gonos 1997, 1998).² We argue that standard employment arrangements are characterized

¹ The February 1995 CPS sample collected data for 106,632 civilian adults of whom the CPS selected 69,002 for the Contingent Work Supplement (CWS). Our analyses are based on the main jobs of the 56,827 respondents who were employed and over age 17. The estimates in Tables 2 to 4 and in Appendix A have been weighted to represent the U.S. population, using weights provided by the Bureau of Labor Statistics.

The CPS collects earnings data from respondents in rotations 4 and 8. Earnings data were also collected for most of the nonstandard workers in the CWS. We constructed an earnings sample of all 21,399 persons over age 17 with earnings data, and constructed a similar sample of the 22,987 respondents with data on union membership. All analyses that include earnings and union membership are based on these samples, weighted (when appropriate) with weights the Bureau of Labor Statistics provided. Other analyses are based on the full sample.

² For example, the Internal Revenue Service treats temporary-help agencies as temporary workers' employers, while many state laws define temporary-help agencies as employment agencies (Gonos 1997, 1998).

Table 1. Characteristics of Standard and Nonstandard Work Arrangements

Type of Work Arrangement	Dimension of Work Arrangements					
	<i>de Jure</i> Employer	<i>de Facto</i> Employer	Assumption of Continued Employment by <i>de Jure</i> Employer	Assumption of Continued Employment by <i>de Facto</i> Employer	Work Directed by	Hours of Work
Standard	Org. A	Org. A	Yes	Yes	Org. A	Full-time
Part-time	Org. A	Org. A	Sometimes	Sometimes	Org. A	Part-time
On-call/day labor	Org. A	Org. A	No	No	Org. A	Full-time or part-time
Temporary-help agency	Agency	Org. A	Sometimes	No	Org. A	Full-time or part-time
Contract company ^a	Contract company	Org. A	Yes	No	Contract company	Full-time or part-time
Independent contractor, self-employed	Self	Client(s)	Yes	No	Self	Full-time or part-time

^a Contract-company employees may have a standard work arrangement with their *de jure* employer (the contract company), but from the point of view of Organization A their work arrangements are nonstandard.

by the exchange of a worker’s labor for monetary compensation from an employer (“Organization A” in Table 1), with work done on a fixed schedule—usually full-time—at the employer’s place of business, under the employer’s control, and with the mutual expectation of continued employment.

It is through the standard employment relationship that most workers earn a living and that the government protects workers from dangerous working conditions through health and safety laws, from exploitation through the Fair Labor Standards Act, from unfair treatment through the National Labor Relations Act and antidiscrimination laws, and from the vicissitudes of unemployment through unemployment insurance. The government also provides additional benefits, such as parental leave and Social Security, through the standard employment relationship.

We use the term “nonstandard work” for employment relations other than standard, full-time jobs, including part-time employment in an otherwise standard work arrangement,³ day labor and on-call work, tempo-

rary-help agency and contract-company employment, independent contracting,⁴ and other self-employment (see Kalleberg et al. 1997 for definitions of these arrangements). These nonstandard arrangements typically depart from standard employment in several respects (see Table 1). First, some lack an employer. This is the case for self-employed persons, independent contractors, consultants, and free-lancers, all of whom bear the risks of their own employment, although

⁴ In the 1995 Contingent Work Supplement, the Bureau of Labor Statistics distinguished between independent contractors who report they are wage and salary employees and those who report they are self-employed. The data from the survey reveal important distinctions between these two groups of independent contractors. Wage-and-salary and self-employed independent contractors differ on the basis of their occupational characteristics as well as the quality of the jobs and the personal characteristics of those who fill them. For example, many wage-and-salary independent contractors are employed in occupations where workers are paid a base salary in addition to some type of commission or additional pay for each product sold or customer served. Self-employed independent contractors are more likely to work in higher status and “professional” occupations than are their wage-and-salary counterparts.

³ Among workers in nonstandard arrangements, we do not distinguish between full-time and part-time employees.

these workers may have many clients who might be said to employ them. Independent contractors, for example, may work at and be paid by a single organization or several organizations, none of whom assumes the legal responsibilities of an employer, such as deducting Social Security taxes or paying unemployment insurance premiums. This fact has led firms to try to evade legal obligations associated with the standard employment relationship (e.g., Callaghan and Hartmann 1991; Johnston 1995). Moreover, while some on-call workers or day laborers—for example, substitute teachers and nurses—work for a single employer (Carré 1992:60), others work out of a hiring hall, and still others are hired from spot markets, like a street corner.

Second, in some nonstandard work arrangements, workers are only weakly attached to their *de jure* (legal) employer in terms of location or administrative control (Pfeffer and Baron 1988). By lobbying state legislatures, the temporary-help industry won the legal status of employer rather than employment agency, thereby freeing its clients from the legal obligations of employers (Gonos 1997). Thus, the *de facto* employers of the employees of temporary-help agencies or contract companies are not their *de jure* employers.

Third, employers do not control how some nonstandard workers do their jobs. For example, independent contractors control their own work; indeed, this is a legal criterion for whether a worker is an independent contractor rather than an employee (Carnevale, Jennings, and Eisenmann 1998:283). Moreover, temporary-help agencies' clients (the *de facto* employers), not the temporary agencies (the *de jure* employers), control the work of temporary-help agency employees.

Fourth, in most nonstandard arrangements workers cannot assume that their employment will continue. Indeed, employers use "on-call" workers, day laborers, workers on fixed-term contracts, temporary-help agency employees, and part-time workers to limit the duration of employment (Pfeffer and Baron 1988). It is this aspect of nonstandard employment that led it to be dubbed "contingent work" (Freedman 1985). Contingent work is generally defined as a job in which there is neither an explicit nor an implicit

contract for long-term employment or in which the minimum hours vary unsystematically (Polivka and Nardone 1989:11). Because contingency *qua* employment insecurity is a dimension of the employment relationship, some full-time jobs may be "contingent."

In 1995, the primary job of nearly 84 million (65 percent) employed Americans over age 17 was a standard full-time job, according to the CPS. Another 17 million worked part-time in standard jobs, and the remaining 20 million worked full-time or part-time in some nonstandard arrangement. Men were more likely than women to hold regular full-time jobs and to be self-employed or independent contractors, and women were overrepresented in regular part-time jobs. Overall, the sexes are similarly distributed across employment relations, with an index of dissimilarity of about 15, but this similarity masks the concentration of men and women in different occupations within the same nonstandard employment arrangements (Kalleberg et al. 1997). Thus, employment in a nonstandard work arrangement may differentially expose the sexes to the risk of a bad job.

BAD JOBS

Critics of nonstandard employment often equate it with poor quality jobs. But only by measuring job quality independently of the type of work arrangement can we determine whether and what kinds of nonstandard employment raise workers' risk of bad jobs. Job quality is an important dimension of stratification in industrial societies, and inequalities in employment relations have drawn considerable attention from sociologists and economists interested in labor markets, class, and the division of labor. For example, Marx ([1867] 1967) contended that employers' ownership and control of the means of production ensured that virtually all jobs would be bad. By the 1940s, U.S. sociologists had replaced this deterministic conception with one that emphasized variation in the prestige of employment (Hodge, Siegel, and Rossi 1964). More recently, sociologists have stressed variability on dimensions such as complexity and physical demands (Cain and Treiman 1981). It is workers, however,

Table 2. Characteristics of Bad Jobs, by Sex and Race/Ethnicity: Current Population Survey, February 1995

Racial/ Ethnic Group ^a	Percent Low Wages			Percent No Health Insurance			Percent No Pension			Mean Number of Bad Job Characteristics		
	Male	Female	Total	Male	Female	Total	Male	Female	Total	Male	Female	Total
White	13.5	††† 23.4	18.1	36.3	††† 49.3	42.3	46.3	††† 52.0	48.9	.95	††† 1.22	1.07
Black	23.3	††† 34.1	29.0	43.5	45.6	44.6	50.0	50.4	50.2	1.25	1.30	1.27
Hispanic	29.3	†† 36.1	32.1	53.8	54.7	54.2	70.4	†† 66.5	66.8	1.47	1.56	1.51
Other	21.5	24.2	22.8	40.1	††† 51.5	45.4	54.1	†† 60.8	57.2	1.21	1.29	1.25
Total	16.1	††† 25.7	20.6	38.7	††† 49.3	43.6	49.1	††† 53.1	51.0	1.03	††† 1.26	1.14
Number of cases	11,607	9,792		28,973	26,411		28,522	26,060		10,982	9,317	

^a All racial/ethnic group comparisons are significantly different at the $p < .001$ level (***), except for the following: *Percent low wages*: black vs. Hispanic (**), black vs. other, and Hispanic vs. other (**) for males; white vs. other and black vs. Hispanic for females; white vs. other (**), black vs. Hispanic (*), and black vs. other (**) for total. *Percent no health insurance*: white vs. other (*) and black vs. other for males; white vs. other, black vs. other (**), and Hispanic vs. other for females; white vs. black (**), white vs. other(*), and black vs. other for total. *Percent no pension*: black vs. other (*) for males; white vs. black and Hispanic vs. other (**) for females; whites vs. blacks for total. *Mean number of bad job characteristics*: black vs. other for males; white vs. black (*), white vs. other, black vs. other, and Hispanic vs. other (*) for females; black vs. other for total.

† $p < .05$ †† $p < .01$ ††† $p < .001$ (two-tailed tests for sex differences)
* $p < .05$ ** $p < .01$ *** $p < .001$ (two-tailed tests for racial/ethnic group differences)

who ultimately judge job quality, and they may consider a job to be bad for many reasons (Jencks, Perma, and Rainwater 1988). Our data allow us to assess three of these—low earnings and the lack of health insurance and pension benefits—that are generally the hallmarks of bad jobs.

Broad consensus exists that wages are a fundamental dimension of job quality (Farber 1997; Jencks et al. 1988). In our analyses, we define a low-wage job as one whose hourly wage is in the bottom quintile in the CPS sample—below \$6.00 an hour (in 1995).

Because workers in the United States obtain health insurance and retirement benefits primarily through employment, jobs that do not offer these benefits may inflict considerable hardship on workers and their families, especially in single-earner households. Thus, our second and third measures of whether a job is bad are whether it provides health insurance and pension benefits. We measure fringe benefits with respondents' reports of whether they have health

insurance and pension coverage through their employer.⁵ We coded self-employed independent contractors and other self-employed workers who purchased fringe benefits as having them. According to the Current Population Survey (CPS), 44 percent of employed persons over age 17 had main jobs that did not provide health benefits, and 51 percent held jobs that did not provide pension benefits. As Table 2 shows, women are less likely to have benefits than men, and people of color—especially Hispanics—are less likely to have benefits than are non-Hispanic whites.

⁵ We measure whether workers have fringe benefits, not whether their employers offer them. In addition to the workers who were insured through their job, another 15.8 percent were offered health insurance and 11.2 percent were offered pension benefits by their employers. Of these, 8.4 and 8.1 percent said health insurance and pension benefits were not offered under conditions or costs that they would or could accept. In addition, 5.5 percent were covered by health insurance through another source.

A COMPOSITE MEASURE OF BAD JOBS

Some scholars contend that bad job characteristics tend to cluster. Dual labor market theorists, for example, posited that primary markets comprise good jobs and secondary markets comprise bad jobs (Piore 1971). By contrast, Smith's ([1776] 1937) hypothesis of compensating differentials holds that pay is positively associated with undesirable working conditions, because workers trade off working conditions and benefits for pay. For example, according to the *CPS Supplement*, in 1995 the majority of professionals working as independent contractors did not receive health insurance (64.3 percent) or pension benefits (52.6 percent), but the high pay and autonomy of independent contracting may have drawn them to work as "free professionals" (Mills 1951). Of course, low-wage jobs are likely to be bad even if they offer another desirable characteristic such as security.

Our measure of the "badness" of a job is its number of bad characteristics, and the three characteristics we use to measure job badness are positively correlated. The correlation between lack of health insurance and the absence of pension benefits is moderate ($r = .48$); both are less strongly correlated with low pay ($r = .33$ for no health insurance and $r = .32$ for no pensions). Thus, employers are not compensating for the absence of benefits with higher pay—a job that is bad on one dimension tends to be bad on others (Hudson 1998).

The main job of about one in seven U.S. workers is bad on all three dimensions (results not shown). Nearly one-quarter (23.1 percent) of workers' main jobs are bad on two dimensions, and almost a quarter (23.9 percent) are bad on one dimension. Just four jobs in 10 have none of these bad qualities.⁶

As Table 2 shows, white men are less likely to have bad jobs than are women of any race or ethnicity or men of color. Of eight sex-race/ethnicity groups, white men are the least likely to hold low-wage jobs, jobs without health insurance, and jobs without pensions. At the other extreme, Hispanic

women are the most likely to be in jobs with low pay and the second most likely, after Hispanic men, to hold jobs that lack pensions. Hispanic men and women are also the least likely to have jobs that provide health insurance.

Except for Hispanics' access to pensions, women's jobs are worse than same-race men's jobs on all three measures, a difference that is not surprising given women's segregation into jobs that pay less than men's jobs and the undervaluation of women's work (England, Reid, and Kilbourne 1996; Reskin 1993). The sex differences in health insurance and pension benefits, however, are limited to married workers (not shown).

LINKING WORK ARRANGEMENTS AND BAD JOBS

We hypothesize that nonstandard work increases workers' risks of bad job characteristics compared with standard employment, and that this difference occurs partly because nonstandard jobs are less secure than standard jobs, less likely to be unionized, and involve less complex work. In addition, because employers use nonstandard arrangements to reduce employment costs, these arrangements should be related to bad job characteristics even after controlling for insecurity, unionization, and complexity. Finally, some nonstandard work arrangements should be associated with worse jobs than others: Independent contractors, other self-employed, and contract-company employees are likely to have fewer bad job characteristics than are temporary employees, on-call workers, and part-time workers.

EMPLOYMENT INSECURITY

Nonstandard work arrangements are more likely than standard jobs to be bad partly because they are less likely to provide employment security. Nonstandard workers often lack explicit or implicit contracts for permanent employment, so their employment, work schedules, and earnings vary unpredictably, depending on employers' needs (duRivage 1986:19). An important reason that employers pay well and provide benefits is to reduce turnover by winning employees'

⁶ Of course, not being bad does not necessarily make a job "good." A "good" job must have other attributes (Jencks et al. 1988).

Table 3. Percentage Distribution of Employment Insecurity, Unionization, and Mean Occupational Complexity, by Type of Work Arrangement and Sex: Current Population Survey, February 1995

Type of Work Arrangement	Total Percent		Percent Uncertain Jobs		Percent Unionized		Mean Occupational Complexity	
	Male	Female	Male	Female	Male	Female	Male	Female
Regular full-time	73.2	††† 64.6	5.4	5.4	21.1	††† 18.3	.002	††† .124
Permanent	69.2	††† 61.1	0.0	0.0	21.0	††† 18.6	-.0004	††† .125
Contingent	3.9	†† 3.5	100.0	100.0	22.5	†† 14.3	.045	.108
Regular part-time	7.7	††† 21.9	15.8	††† 11.8	8.6	8.2	-.365	††† -.164
On-call and day labor	1.4	†† 1.7	47.9	†† 37.5	27.3	††† 10.8	-.459	††† -.022
Temporary-help agency	.8	†† 1.1	78.2	76.0	3.3	2.2	-.581	††† -.253
Contract company	1.7	††† .9	17.0	14.8	16.2	††† 4.1	.004	.060
Independent contractor, wage-and-salary worker	.9	.9	20.9	24.8	5.9	3.2	.409	††† -.041
Independent contractor, self-employed	7.6	††† 3.9	7.0	†† 10.8	NA	NA	.299	††† .116
Other self-employed	6.7	††† 5.0	6.3	††† 10.5	NA	NA	.458	††† .152
Total	100.0	100.0	7.9	††† 8.8	16.8	††† 14.0	.020	††† .053
Number of cases	29,772	27,055	29,772	27,055	12,663	10,324	29,772	27,055

†*p* < .05 ††*p* < .01 †††*p* < .001 (two-tailed tests for sex differences)

loyalty. However, employers may not offer these incentives to workers they don't plan to retain. Thus, employers are less likely to offer benefits and more likely to pay low wages to workers whose future employment is uncertain (Hudson 1998).

Table 3 shows that uncertainty is more common in nonstandard than in standard jobs.⁷ Overall, between 8 and 9 percent of respondents had jobs that were insecure. In almost every nonstandard work arrangement, both sexes' jobs were substantially less secure than those of regular full-time workers: Only 5.4 percent of men and women in regular full-time jobs reported that their jobs were of uncertain duration. Only

⁷ We classified the jobs of self-employed independent contractors and other self-employed workers as insecure if they expected them to last no more than one year. We classified wage-and-salaried employees as having insecure jobs if they described them as temporary, were unsure if they were temporary, could not work for their employer as long as they wished, or were unsure how long they could work for their present employer. We did not code jobs that were temporary for noneconomic reasons as contingent.

male self-employed independent contractors and other self-employed workers reported a comparable level of job security. By contrast, over three-quarters of temporary-help agency employees reported that their job duration was uncertain. Sex was not consistently related to exposure to employment uncertainty: Men who worked part-time, through temporary-help agencies, and were on call were more likely than women in these arrangements to have uncertain jobs, while self-employed women and independent contractors were more likely than their male counterparts to describe their job's duration as uncertain, reflecting the different economic situations of self-employed men and women. In sum, since employment insecurity is common in nonstandard employment, controlling for it should reduce the effect of nonstandard employment relations on the number of bad job characteristics.

UNIONIZATION

Through collective bargaining, unions have improved the quality of workers' jobs. However, the right to bargain collectively pro-

tected by the National Labor Relations Act is meaningless for workers whose *de jure* employer is not their *de facto* employer (Hiatt 1995:742), so working for temporary-help agencies and contract companies effectively excludes workers from union representation (Gonos 1998). A contracting company—the *de facto* employer—can cancel the contract if workers unionize, often leaving them jobless and without legal recourse (Hiatt 1995:753). If unionization reduces workers' exposure to bad job characteristics and standard work arrangements are more likely than nonstandard arrangements to be unionized, then unionization should help account for the association between nonstandard work and bad job characteristics.

The CPS workers in regular full-time jobs were more likely to be unionized than were workers in most nonstandard arrangements (Table 3). The high unionization rate among male on-call/day laborers reflects the high rates of unionization by truck drivers, electricians, carpenters, plumbers, pipefitters, and steamfitters in these nonstandard work arrangements.

OCCUPATIONAL COMPLEXITY

Employers try to retain skilled workers by offering higher wages and fringe benefits. Thus, we expect jobs that involve more skill to be better than those that require less skill. Because highly skilled jobs tend to be complex, we use the *complexity* of a worker's occupation as a proxy for their job skill. Our measure of occupational complexity is based on several variables from the *Dictionary of Occupational Titles*.⁸

According to Table 3, male wage-and-salaried and self-employed independent contractors, and male and female "other" self-employed persons, held more complex jobs than workers in other arrangements. Contract-company employees had lower skill

levels than self-employed independent contractors and other self-employed workers, but higher levels than temporary-help agency employees, on-call workers and day laborers, and part-time workers. These findings are consistent with our assumption that independent contractors and the self-employed are most likely to control their own work (see Table 1). If higher occupational complexity reduces workers' exposure to bad job characteristics, temporary employees, on-call workers/day laborers, and part-time employees should have more bad job characteristics than workers in standard work arrangements or those in the other nonstandard work arrangements.

CUTTING EMPLOYMENT COSTS

Establishments often hire nonstandard workers to reduce employment costs, such as transaction and administrative costs and, especially, benefit costs, which constitute between 30 and 40 percent of total compensation (Cascio 1992). In fact, avoiding fringe benefit costs is a major reason employers use nonstandard work arrangements (Blank 1998; Houseman 1997). Part-time jobs are less likely than full-time jobs to provide health insurance, unemployment insurance (Tilly 1996), and pensions (Belous 1989). Under the Employment Retirement Security Act, employers can exempt from pension plans workers employed fewer than 1,000 hours a year (duRivage 1986:23), and independent contractors' and free-lancers' self-employed status exempts their clients from contributing to Social Security or unemployment insurance (Callaghan and Hartmann 1991:26). Temporary-help agencies generally offer no benefits or limit benefits to workers who have worked a specified number of hours (duRivage 1986:22). Regulations enacted in 1985 allow the federal government to exempt health and retirement benefits (except Social Security) from temporary jobs, bringing government practices in line with the private sector (Appelbaum 1987:274, 281). Whether nonstandard workers qualify for unemployment compensation depends on their state of residence and their hours of work (duRivage 1986:24). In sum, nonstandard work reduces workers' likelihood of receiving benefits. Because we have

⁸ Our measure is based on general educational development (GED), specific vocational preparation (SVP), complexity in dealing with data and people, verbal and numerical aptitude, and the reasoning ability an occupation requires. After recoding GED and SVP so high scores indicated more complex jobs, we standardized and averaged the seven indicators. The scale has a Cronbach's alpha of .97.

Table 4. Percentage Distribution of Bad Job Characteristics and Mean Number of Bad Job Characteristics, by Type of Work Arrangement and Sex: Current Population Survey, February 1995

Type of Work Arrangement	Percent Low Wages		Percent No Health Insurance		Percent No Pensions		Mean Number of Bad Job Characteristics	
	Male	Female	Male	Female	Male	Female	Male	Female
Regular full-time	11.2 †††	16.0	29.0 †††	33.1	41.3	40.3	.82 ††	.87
Permanent	11.0 †††	15.6	28.4 †††	32.4	40.5	39.5	.81 ††	.86
Contingent	15.9 †	22.0	38.9 †	44.4	54.2	54.1	1.08	1.22
Regular part-time	52.6 †	48.6	83.5 †	81.5	88.0 †††	79.1	2.21 ††	2.11
On-call and day labor	31.1	40.7	70.9 †††	88.5	81.4	81.8	1.81 ††	2.09
Temporary-help agency	44.2	35.2	97.2 †	93.5	96.9	98.3	2.39	2.27
Contract company	16.0 ††	31.2	49.9 †††	66.7	65.7 †	73.3	1.31 ††	1.70
Independent contractor, wage-and-salary worker	14.0 ††	29.9	68.3 †††	90.2	78.2 †††	91.5	1.61 †††	2.09
Independent contractor, self-employed	16.2 †††	34.2	61.0 †††	75.2	62.7	64.6	1.38 †††	1.72
Other self-employed	23.4 †††	45.2	47.3 †††	63.6	54.3 †††	62.9	1.23 †††	1.71
Total	16.1 †††	25.7	38.7 †††	49.3	49.1 †††	53.1	1.03 †††	1.26
Number of cases	11,607	9,792	28,973	26,411	28,522	26,060	10,982	9,317

†*p* < .05 ††*p* < .01 †††*p* < .001 (two-tailed tests for sex differences)

no measure of the employment costs for those in the CPS sample, we can test this explanation only indirectly by assessing whether nonstandard work arrangements are associated with more bad job characteristics, net of employment insecurity, unionization, and job complexity.

WORK ARRANGEMENTS AND BAD JOBS: BIVARIATE ASSOCIATIONS

The relationship between work arrangements and bad job characteristics for men and women is summarized in Table 4. About one in nine men and one in six women in regular full-time jobs are in the bottom wage quintile. Other self-employment substantially exacerbates women's vulnerability to lower pay. Among permanent full-time workers, the sex difference in the likelihood of a low-wage job is 4.6 percentage points. Among other self-employed, it is 21.8 percentage points. The nonstandard relationships whose incumbents are at disproportionate risk of low pay include part-time employment, on-call and day labor, temporary-

help employment, and other self-employment for both sexes; and contract work and both types of independent contracting for women.

Almost 40 percent of men and 50 percent of women do not receive health insurance from their main jobs, and nearly half the men and over half the women receive no pension benefits. As we expected, workers in standard work arrangements are most likely to obtain these benefits: Less than one-third lack employer-provided health insurance, and about 40 percent lack pension benefits. In no other work arrangement are coverage rates this high—over 90 percent of employees of temporary-help agencies, 80 percent of men and women in regular part-time jobs, and over 70 percent of day laborers and on-call workers do not receive these benefits.

Self-employed independent contractors, other self-employed, and contract employees fall in between part-timers, temporary-help agency employees, and on-call workers/day laborers, on the one hand, and standard full-time employees, on the other, in their access

to benefits. The higher likelihood of these benefits among male than female independent contractors and self-employed workers partly reflects the larger scale of their enterprises. Self-employed women are more likely than self-employed men to have a shoestring operation with no employees (Spalter-Roth, Hartmann, and Shaw 1993).⁹ Among contract workers and on-call/day laborers, men are more likely than women to have health benefits. Male contract-company workers are more likely than their female counterparts to have pension benefits. These sex differences almost certainly stem from occupational differences. The modal occupation for male contract worker is systems analyst, while that for female contract workers is nurse's aide; the modal occupation for male day laborers and on-call workers is truck driver, while for women it is substitute teacher.

Table 4 presents the mean number of bad job characteristics by work arrangement. The jobs of workers in standard arrangements have fewer bad characteristics than those of workers in nonstandard arrangements. Moreover, self-employed workers and contract employees have jobs with fewer bad characteristics than regular part-timers, temporary-help agency employees, and on-call workers/day laborers. Women average more bad job characteristics than men in five of the seven nonstandard arrangements, and the sex difference is substantially greater in nonstandard work than in regular full-time jobs. However, the number of bad job characteristics of male and female temps—the work arrangement with the most bad characteristics for each sex—do not differ significantly. Moreover, among part-time workers—the work arrangement associated with the second highest number of bad characteristics—men's jobs have significantly more bad characteristics than women's jobs.

⁹ Male wage-and-salaried independent contractors, incorporated independent contractors, and other incorporated self-employed persons averaged 18 employees, while women in these work arrangements averaged 10. The logarithm of independent contractors' or other self-employed workers' number of employees correlates .21 with whether they had health insurance and .16 with whether they had pension plans.

These bivariate results are consistent with workers in nonstandard employment encountering bad job characteristics more often than do workers in standard arrangements. Moreover, nonstandard work arrangements in which workers exercise greater control over their work (i.e., self-employed, independent contractors, and contract employees) are associated with better jobs than the other kinds of nonstandard work arrangements.

MULTIVARIATE ANALYSIS

OTHER EXPLANATORY VARIABLES

Before testing our hypotheses about the effect of work arrangements on bad jobs, we must consider other factors that affect workers' risk of bad jobs. Workers presumably seek jobs with as few bad characteristics as possible, but their ability to obtain better jobs depends on their relative power, as individuals and collectively, to obtain advantaged positions in the stratification system (Form and Huber 1976; Kalleberg, Wallace, and Althausen 1981; Morse 1969; Reskin 1991). We have discussed several sources of workers' power, such as unions and skills that are associated with different work arrangements. In addition, a worker's *education* signals her/his degree of skill and hence bargaining power. Because employers generally prefer more educated workers, the latter have better options than less educated workers. To capture respondents' educational attainment and education's credentialing effect, we measure education with a set of dummy variables for the highest level of schooling the respondent completed. To control for the fact that students' jobs are typically temporary, we include a binary variable indicating whether workers aged 18 to 24 are *currently in school* (coded 1).

A worker's *family status* may also be related to the quality of her or his job. The fact that women's role as children's primary caretaker has disproportionately concentrated women in part-time work may contribute to this association. We take this into account with measures indicating whether a worker has children (coded 1) and whether she or he is unmarried or married with a stay-at-home spouse compared with married

with an employed spouse (the reference category).¹⁰

Because older workers have more labor force experience and more seniority with their employers, their jobs should have fewer bad characteristics than those of younger workers. Beyond a certain age, however, older workers lose bargaining power because they lack employment alternatives. To model this, we use an age-squared term to allow the effect of age on number of bad job characteristics to decline as workers age.

To control for factors affecting job quality, such as the supply of and demand for various kinds of labor, characteristics of product markets, and so on, our models include broad *occupation* and *industry* categories. Net of unionization, occupations differ in their market power and hence in incumbents' ability to garner high wages and fringe benefits. Industries vary in employers' need to minimize their wage commitments and in their ability to invest in and capitalize on workers' training (Knoke and Kalleberg 1994). We control for economic context by including measures of urban-rural status (suburban is the reference category) and region (Northeast is the reference category). Appendix A presents descriptive statistics for our explanatory variables. To control for selection bias resulting from excluding the nonemployed and the unemployed from the sample, we also estimated our models including inverse Mill's ratios (results not shown). None of the coefficients for selection was statistically significant.

MODEL

Our dependent variable, *the number of bad job characteristics*, is a count that ranges from 0 to 3. Count variables typically have a Poisson or a negative binomial distribution (Long 1997). Because Poisson regression models assume that the conditional mean is

¹⁰ To allow a spouse who can share childcare responsibilities to affect a worker's ability to obtain a job with fewer bad characteristics, we included interaction terms for children by marital status and spouse's employment status. Neither term was significant, so neither appears in the final equation.

equivalent to the conditional variance—a fairly restrictive assumption—we used negative binomial regression to assess the determinants of the “badness” of workers' jobs.¹¹ To facilitate interpretation, we converted the negative binomial regression coefficients to estimates of the percentage change in the number of bad job characteristics associated with a one-unit increase in the value of the independent variable, net of the other variables in the model. We computed the percentage change using the following transformation:

$$[(\exp b) - 1]100.$$

For example, holding all other variables constant, black men's jobs have 9.1 percent more bad characteristics than white men's, the reference category (see column 3, Table 5). Unlike the change in the actual count of bad job characteristics, the percentage change in bad job characteristics associated with a given independent variable does not depend on the value of other variables in the model. As in logistic regression, however, the impact of any independent variable on the expected count depends on the values of the other independent variables. For example, we would expect a white, single, childless woman of average age who lives in the suburbs in the Northeast, has a high school diploma, works as a manager in manufacturing, has a regular full-time permanent nonunion job, and whose occupation is of average complexity to have .74 bad job characteristics.¹² Employment by a tempo-

¹¹ Although some researchers use OLS regression to estimate count variables, count distributions are likely to violate the OLS assumptions that the disturbance terms are homoskedastic and the error term is uncorrelated with the independent variables, especially when the count on the dependent variable is low. See Long (1997, chap. 8) for a discussion of the differences between Poisson and negative binomial regression models.

¹² Because we estimated the negative binomial regression coefficients using *Stata* and unweighted data, the means of the continuous variables we use in this example differ slightly from the means shown in Appendix A. The unweighted means for the continuous variables for women are 39.68 for age, and .06 for occupational complexity; for men these values are 40.65 for age, and .11 for occupational complexity.

rary-help agency would raise her predicted count of bad job characteristics by 103 percent to a total of 1.5 bad characteristics.¹³ A similar man in a regular full-time job could expect to have .64 bad job characteristics compared to 1.2 for a man in a temporary job. The effect on the count of bad job characteristics of having a temporary job would differ if workers' values differed on any of the independent variables.

RESULTS

Consistent with our main hypothesis, every type of nonstandard employment is associated with more bad job characteristics than is standard full-time employment. Model 1 in Table 5 presents estimates for both males

¹³ The equation for the expected number of bad job characteristics for the regular full-time worker in this example is:

$$\begin{aligned} E\{\text{Count}\} &= \exp[\text{intercept} + \beta(\text{age}) + \beta(\text{age}^2) \\ &\quad + \beta(\text{complexity})] \\ &= \exp[.5126 - .0358(\text{age}) \\ &\quad + .00035(\text{age}^2) \\ &\quad - .0890(\text{complexity})] \\ &= .7419. \end{aligned}$$

The equation for a temporary-help agency (THA) worker is:

$$\begin{aligned} E\{\text{Count}\} &= \exp[\text{intercept} + \beta(\text{age}) + \beta(\text{age}^2) \\ &\quad + \beta(\text{complexity}) + \beta(\text{THA})] \\ &= \exp[.5126 - .0358(\text{age}) \\ &\quad + .00035(\text{age}^2) \\ &\quad - .0890(\text{complexity}) \\ &\quad + .7101(\text{THA})] \\ &= 1.5093. \end{aligned}$$

The negative binomial regression coefficients for any independent variable can be obtained from the percentage changes in Table 5 using the following formula:

$$\ln [(\text{percentage}/100) + 1].$$

(The standard errors from these negative binomial regressions are available from the authors.) The intercepts are .5126 for women and .5251 for men. The unweighted means of the continuous variables given above and negative binomial regression coefficients—which can be computed with the formula above using the percentage changes in Table 5—can be used to compute the expected count for any combination of independent variables.

and females of the percentage change in the number of bad job characteristics based on equations containing our measures of work arrangements and also including measures of workers' personal characteristics. Model 2 in Table 5 adds measures of employment insecurity, unionization, and skill plus dummy variables for occupation and industry.

In Model 1, as we expected, the percentage changes for all nonstandard work arrangements for both sexes are positive and statistically significant, indicating that nonstandard employment arrangements have more bad job characteristics than do standard full-time arrangements, the reference category.¹⁴ Employment in nonstandard work arrangements increases the risk of bad job characteristics quite substantially. Arrangements that approximate the standard employment relationship (such as contract work) or that give workers a relatively high degree of control over their work (such as the two forms of self-employment) exert smaller effects on the number of bad job characteristics. By contrast, people who work as temporaries and regular part-time workers and women on-call or day laborers are substantially worse off relative to workers in standard work arrangements.

Model 2 in Table 5 shows that employment insecurity, unionization, and job complexity are all significantly related to the percentage change in the number of bad job characteristics. This finding, combined with the finding that these mechanisms are associated with the nonstandard work arrangements (see Table 3), supports our argument that insecurity, unionization, and complexity help to explain the association between nonstandard arrangements and bad job characteristics. Women and men who hold an insecure job experience about 13 percent and 18 percent, respectively, more bad job characteristics, controlling for all of the other variables in our model (see Model 2 for each sex). In addition, men and women covered by collective bargaining agreements have jobs with half as many (52 to 55 percent)

¹⁴ The analyses in Tables 5 to 7 include a dummy variable denoting workers who are in regular, full-time jobs of uncertain duration, so our reference category (i.e., regular, full-time, noncontingent jobs) corresponds to our conception of the standard employment relation.

Table 5. Percentage Change in Bad Job Characteristics Calculated from Negative Binomial Regressions of Number of Bad Job Characteristics on Work Arrangements and Other Explanatory Variables, by Sex: Current Population Survey, February 1995

Independent Variable	Model 1		Model 2	
	Male	Female	Male	Female
	Percentage Change	Percentage Change	Percentage Change	Percentage Change
<i>Work Arrangement</i>				
[Standard]				
Regular contingent	49.7***	64.9***	26.0***	42.5***
Regular part-time	117.4***	126.8***	75.8***	87.1***
On-call and day labor	82.1***	††† 135.2***	56.0***	††† 98.4***
Temporary-help agency	138.8***	142.4***	89.6***	103.4***
Contract company	52.0***	†† 92.0***	39.1***	†† 71.3***
Independent contractor, wage and salary worker	128.5***	155.5***	90.9***	106.5***
Independent contractor, self-employed	91.8***	†† 118.6***	54.0***	† 68.0***
Self-employed, other	72.9***	††† 108.1***	39.8***	†† 62.0***
Employment insecurity	—	—	17.5***	12.8***
Unionization	—	—	-54.5***	-51.6***
Occupational complexity	—	—	-15.1***	† -8.5***
<i>Race/Ethnicity</i>				
[White]				
Black	12.0**	3.2	9.1*	3.3
Hispanic	13.2**	2.6	6.9	3.0
Other	9.3	-1.2	7.9	-1.4
Born outside the U.S.	22.5***	16.6***	13.8***	11.4**
<i>Family Status</i>				
[Married, spouse employed]				
Single	9.1***	†† -0.6	5.4*	†† -4.0
Married, spouse unemployed	-4.5	1.3	-3.9	-1.8
Children	-3.0	†† 5.7**	-1.9	†† 6.4**
<i>Education</i>				
[High school grad]				
Less than high school grad	18.0***	16.1***	9.4**	6.0
Some college	-2.5	-8.4***	1.0	-3.2
Associate degree	-15.1***	-19.5***	-8.9*	-12.1***
BA degree	-27.8***	-27.6***	-17.8***	-13.2***
Advanced degree	-35.2***	-41.2***	-19.0***	-22.4***
Attending school	-3.3	1.5	-3.0	2.3
Age	-5.9***	† -4.5***	-4.2***	-3.5***
(Age) ²	.1***	† .0***	.0***	.0***

(Continued on next page)

(Table 5 continued)

Independent Variable	Model 1		Model 2	
	Male	Female	Male	Female
	Percentage Change	Percentage Change	Percentage Change	Percentage Change
<i>City Type and Region</i>				
[Suburban]				
Urban	8.8***	1.7	8.1**	2.4
Rural	13.5***	15.5***	7.5**	11.5***
City type not identified	6.8*	5.9*	3.8	4.0
[Northeast]				
Midwest	.2	4.2	-1.2	1.9
South	9.9***	8.5**	6.1*	4.6
West	5.3	3.1	3.2	2.5
<i>Occupation</i>				
[Manager]				
Professional	—	—	10.3*	3.9
Technical	—	—	-6.0	-8.2
Sales	—	—	9.9*	14.4**
Clerical	—	—	.9	3.3
Private household	—	—	15.6	24.7
Protective services	—	—	20.8*	12.2
Other service occupations	—	—	19.4***	27.6***
Craft	—	—	3.8	8.8
Machine operatives	—	—	8.6	20.2**
Transport operatives	—	—	15.1*	9.8
Laborers	—	—	2.4	17.0
Farm and forestry	—	—	32.9**	16.8
<i>Industry</i>				
[Manufacturing]				
Agriculture, forestry, fishery	—	—	22.8*	36.9***
Mining	—	—	-8.2	-26.2
Construction	—	—	39.6***	† 16.4
Transportation	—	—	7.1	.7
Wholesale trade	—	—	13.1*	14.8*
Retail trade	—	—	43.8***	35.9***
Finance, real estate	—	—	16.8**	† -2.0
Private households	—	—	20.8	18.0
Business, repair services	—	—	42.0***	† 23.3***
Personal services	—	—	40.2***	34.5***
Entertainment, recreation	—	—	42.3***	39.2***
Professional services	—	—	25.8***	15.8***
Public administration	—	—	-37.9***	-23.2***
<i>p</i> -value of Alpha test	.980	.980	.984	.971
Pseudo R-square	.095	.105	.130	.135
Number of cases	10,977	9,286	10,977	9,286

† *p* < .05 †† *p* < .01 ††† *p* < .001 (two-tailed tests for sex differences)
* *p* < .05 ** *p* < .01 *** *p* < .001 (two-tailed tests)

bad job characteristics as workers who are not covered. The complexity of a person's occupation is negatively associated with its number of bad characteristics. Although the effects of unionization and employment insecurity do not differ statistically for the sexes, occupational complexity has a significantly greater negative effect on men's number of bad job characteristics.

Also consistent with our expectations, nonstandard work arrangements are significantly associated with more bad characteristics than are standard work arrangements, net of the effects of insecurity, unionization, complexity, occupation, and industry (see Model 2 for each sex in Table 5). However, the controls substantially reduce the effects of several of the nonstandard arrangements. Taking into account occupational complexity, employment uncertainty, and unionization, as well as occupation and industry reduces the effects of both types of self-employment on men's undesirable job characteristics by at least 41 percent, and reduces the effects of both forms of self-employment on women's bad job characteristics by 43 percent. The smallest reductions in the percentage of bad job characteristics associated with these controls are 22 and 25 percent for female and male contract employees.

Thus, employment in nonstandard work arrangements exposes workers to significantly worse jobs than does employment in standard full-time jobs, net of worker's age, education, occupation, and industry. In addition, for both sexes, temporary-help agency employment, regular part-time work, wage-and-salaried independent contracting, and (for women) on-call/day labor are associated with worse jobs than are the other nonstandard work arrangements. Importantly, the effects of the nonstandard work arrangements relative to standard work arrangements generally exceed those of the other variables in the equation.

Overall, women's jobs have 22 percent more bad characteristics than do men's jobs (1.26 versus 1.03; see Table 2). Net of the effects of nonstandard working arrangements and the other variables in Table 5, women have 13.6 percent more bad job characteristics than men ($p < .001$, results not shown). Although the percentage changes associated with temporary work, part-time

work, and wage-and-salaried independent contracting are greater for women than for men, these sex differences do not reach conventional levels of significance. However, working on call or as a day laborer, employment with a contract company, self-employed independent contracting, and other self-employment increase women's exposure to negative job qualities relative to their counterparts in standard jobs significantly more than men's (see the last column in Table 5). These greater penalties for women are *not* artifacts of women having better "standard" jobs than men (see Table 4). Thus, nonstandard employment exacerbates sex inequality in the exposure to bad job characteristics.

Black men experience significantly more bad job characteristics than do white men; the other race differences were not significant at the $p < .05$ level in the full models. Nativity is significantly associated with workers' number of bad job characteristics: Immigrants of both sexes are exposed to 11 to 14 percent more bad job characteristics than same-sex native-born workers.

Single men have significantly more negative job characteristics than do married men, mirroring their earnings disadvantage compared with married men (Korenman and Neumark 1991). The job quality of single women does not differ significantly from that of married women. For married women and men, access to a partner's earnings does not appear to matter. Having children has no significant effect on how many bad characteristics men's jobs have, but is associated with significantly more bad job characteristics for women. The concentration of mothers in part-time work is likely to help explain this effect.

Our control variables operate as expected. Higher education reduces bad job characteristics for both sexes, and men with less than a high school diploma have worse jobs than do high school graduates. Age reduces men's and women's bad job characteristics, but at a slightly diminishing rate over the course of their careers.

Workers in manufacturing have fewer bad job characteristics than workers in most other industries; only workers in public administration have significantly fewer bad job characteristics than manufacturing workers.

Table 6. Odds Ratios from Logistic Regressions Predicting Bad Job Characteristics from Work Arrangements and Other Independent Variables, by Sex: Current Population Survey, February 1995

Independent Variable	Low Wage		No Health Insurance		No Pension	
	Males	Females	Males	Females	Males	Females
<i>Work Arrangement</i>						
[Standard]						
Regular contingent	1.44*	1.42*	1.10	1.29	1.01	1.70***
Regular part-time	3.58***	3.20***	6.10***	6.49***	4.41***	4.64***
On-call and day laborer	2.18***	2.89***	3.54***	12.89***	3.48***	6.77***
Temporary-help agency	3.01***	2.07***	27.53***	19.10***	11.92***	33.20***
Contract company	1.36	1.74**	1.92	3.38***	1.99***	3.66***
Independent contractor, wage-and-salary worker	1.64*	2.12***	5.28***	12.79***	6.38***	14.63***
Independent contractor, self-employed	2.36***	3.07***	3.02***	4.38***	1.82***	2.08***
Self-employed, other	2.84***	4.35***	1.79***	2.40***	1.47***	1.79***
Employment insecurity	1.21	1.32**	1.96***	2.06***	2.79***	1.84***
Unionization	.34***	.33***	.24***	.31***	.21***	.21***
Occupational complexity	.61***	.65***	.58***	.82**	.71***	.83**
Number of cases	11,532	9,694	12,312	10,052	12,076	9,920

Note: Equations include controls for variables in Model 2 reported in Table 5.

* $p < .05$ ** $p < .01$ *** $p < .001$ (two-tailed tests)

Jobs in some broad occupational categories have more bad job characteristics than managers, the reference category. Women operatives, sales workers, and service workers—nearly one-third of all women (see Appendix A)—experience significantly more bad job characteristics than female managers do. Finally, men in suburban areas have better jobs than those in urban or rural areas, and women in suburban areas have better jobs than those in rural areas.

EFFECTS OF WORK ARRANGEMENTS ON WAGES AND FRINGE BENEFITS

Does the fact that our composite measure gives equal weight to the three bad job characteristics affect our estimates of the impact of work arrangements? Separate analyses of the three measures of bad jobs (Table 6) suggest not. Workers in standard arrangements are significantly more likely to have both health insurance and pension benefits than are workers in each of the nonstandard arrangements, net of personal characteristics,

employment insecurity, unionization, skill, occupation, and industry. Contract employees and the self-employed are most similar to workers in standard arrangements and are more likely to have health insurance and pension benefits than temporary-help agency employees, regular part-time workers, and (especially women) on-call/day laborers.¹⁵

Workers in standard employment relations are also less likely than those in nonstandard work arrangements to have a low-wage job, with the exception of male contract employees (Table 6). The exception conforms to our recognition that contract employees may have standard employment arrangements with the contract company, so their employment is nonstandard only in its location (see note to Table 1). The difference between standard work arrangements and

¹⁵ When we reestimated the fringe benefit models in Table 6 using whether the worker was offered the benefit as the dependent variable (see Footnote 5), we obtained essentially the same results.

Table 7. Percentage Change from OLS Regression of (log) Wages and Odds Ratios from Logistic Regressions of High Wages on Work Arrangements and Other Independent Variables, by Sex: Current Population Survey, February 1995

Independent Variable	Percentage Change in (log) Wages ^a		Odds Ratios for High Wages ^b	
	Male	Female	Male	Female
<i>Work Arrangement</i>				
[Standard]				
Regular contingent	-2.0	1.2	.86	1.41
Regular part-time	-17.7***	-11.9***	.49***	.86
On-call and day laborer	-6.2	-12.0***	.88	.82
Temporary-help agency	-14.8**	-6.9	.68	.99
Contract company	4.3	.2	1.37*	1.76**
Independent contractor, wage and salary worker	1.3	1.4	.97	1.56*
Independent contractor, self-employed	-4.1*	-12.8***	1.06	1.93***
Self-employed, other	-9.8***	-19.7***	1.03	1.50***
R ² (adjusted)	.25	.27	—	—
Number of cases	11,383	9,433	11,532	9,694

Note: Equations include controls for variables in Model 2 reported in Table 5.

^a Percentage change in wages from OLS regression.

^b Odds ratios from logistic regression.

p* < .05 *p* < .01 ****p* < .001 (two-tailed tests)

wage-and-salaried independent contracting is also relatively small. Regular part-time workers, female self-employed persons, male temporary-help agency employees, and female on-call/day laborers are especially likely to earn low wages.

Taken together, the results in Table 6 reinforce our conclusion that standard arrangements provide better jobs than do nonstandard arrangements. Moreover, these results are consistent with our expectation that contract employees have better jobs than temporary-help agency employees, part-time workers, and on-call workers/day laborers. Both types of self-employed workers are also relatively less likely to have jobs with no fringe benefits.

Table 7 presents two additional analyses of the effects of work arrangements on wages. The first set of columns presents OLS estimates of the effects of nonstandard work arrangements on (log) hourly wages. The second set regresses the odds that workers in specific nonstandard work arrangements are in the top fifth of the wage distri-

bution (i.e., their job pays at least \$17.32 per hour) on nonstandard employment and the control variables.

The first two columns further demonstrate the disadvantages associated with regular part-time work and self-employment (both categories): Women and men in these nonstandard arrangements earn significantly lower hourly wages than do workers in standard arrangements. Women working as on-call/day laborers also earn lower hourly wages than do women in standard arrangements. Temporary-help agency employment is more costly for men than for women (the coefficient for women is not significant) relative to their same-sex counterparts in standard jobs as women's wages in standard jobs are already low.

The logged hourly wages of contract-company employees and wage-and-salaried independent contractors do not differ from those of standard employees, net of the other variables, reinforcing our conclusion that contract employees resemble employees in standard work arrangements as far as wages

are concerned. Indeed, contract employees have a higher probability than employees in standard work arrangements to be in the top-quintile of wages. This is also true for both types of self-employed women.

The results in Table 7 underscore the heterogeneity of nonstandard work arrangements. In particular, contract-company employees, wage-and-salaried independent contractors, and both types of self-employment include significant numbers of both high- and low-wage jobs. By contrast, temporary-help agency employment, on-call/day labor, and part-time work are generally associated with worse characteristics than are regular full-time jobs.

CONCLUSIONS AND IMPLICATIONS

Nonstandard work arrangements have increased in the United States over the past two decades. This is especially true for temporary employment, which has grown by about 11 percent per year since 1972 (Gonos 1997; Segal and Sullivan 1997). Part-time employment has grown from 16.4 percent of the labor force in 1970 to 18 percent in 1990, mostly through the growth of involuntary part-time employment (Tilly 1996). For example, in the mid-1980s almost all bank tellers worked full-time; by the early 1990s, 60 percent were employed part-time (Keltner and Finegold 1996). An increase in the number of persons reporting income only as self-employed or independent contractors suggests growth in independent contracting (Callaghan and Hartmann 1991).

Debates over the quality of nonstandard jobs are often related to discussions about why nonstandard work arrangements have increased. If nonstandard jobs are systematically worse than standard jobs, then the growth in nonstandard work is more likely to reflect employers' rather than workers' preferences (Blank 1998; Pfeffer and Baron 1988). Distinguishing analytically two key concepts—the quality of jobs associated with various employment relations from whether an arrangement is nonstandard—helps to clarify these debates over nonstandard jobs in the United States.

Much of the research on work and labor markets has argued that the firm is the cen-

tral context for understanding labor market structures (Baron and Bielby 1980). Our results underscore the importance of also taking into account the type of employment relationship for understanding the nature and outcomes of labor market structures. It is a very different matter to be employed on a full-time basis by a Fortune 500 firm than to work in the same firm as a temporary, contract, or on-call worker. In some cases, the structure of employment relations overrides firm and sector effects on workers' exposure to good or bad jobs. Thus, we found that every nonstandard work arrangement we examined is more likely to be associated with bad job characteristics than standard work arrangements. At the same time, we found that some workers in certain nonstandard work arrangements—particularly self-employment (both independent contracting and other forms of self-employment) and contract-company employment—had jobs that were not all that bad. While workers in these nonstandard arrangements were less likely to have fringe benefits than workers in standard arrangements, many earned higher wages than regular full-time workers in standard jobs. Moreover, relatively few workers in these particular nonstandard arrangements expressed a preference for standard jobs (Kalleberg et al. 1997). The heterogeneity within some nonstandard work arrangements is seen in the overrepresentation of some workers—such as self-employed women—in both the low- and high-wage groups.

On the other hand, male and female temporary-help agency employees, on-call workers and day laborers, and part-time workers are consistently more likely than regular full-time workers—and workers in the other nonstandard arrangements—to have low pay and to lack insurance and pension benefits. Moreover, most workers in these arrangements (especially temporary-help agency employees and on-call workers and day laborers) prefer standard, full-time employment (Kalleberg et al. 1997). In view of these findings, the explosive growth of the temporary-help industry, in particular, makes the strong negative effect of employment in temporary-help agencies on job quality a matter of concern.

Nonstandard work arrangements alter the power dynamics between employers and

employees. For example, temporary work and contract work, arrangements in which the legal (*de jure*) employer differs from the actual (*de facto*) employer to whom workers provide services, expose workers to fundamentally different power dynamics than do standard, bilateral employment relations. The separation of the “legal employer” from the “supervising employer” subjects workers to two masters, thereby creating triangular employment relations (Córdova 1986). Contract and temporary employees are subject to the demands of their *de facto* employers, but without the hope of security and advancement with these employers that “real” employees can entertain. Thus, the pay and benefits of nonstandard workers are not subject to the incentive structures that their *de facto* employer provides to its “real” employees. At the same time, the *de jure* employer is ill situated to evaluate their employees’ performance and has minimal stake in doing so; these workers (particularly temporary-help agency employees) have little hope of improving their position with their *de jure* employer by working hard or performing well.

The triangular nature of some nonstandard employment arrangements in which the *de jure* and *de facto* employers differ also limits these workers’ ability to reduce the power imbalance with either employer through collective bargaining. For example, temp workers do not have the sustained contact with their *de jure* coworkers necessary to organize collectively to demand better benefits from the temporary-help agencies that nominally employ them. Moreover, their transient status and the fact that their *de facto* coworkers do not share their marginal status prevent collective action with their *de facto* coworkers. Contract workers are also poorly situated to enhance their power through collective bargaining because the National Labor Relations Act does not prohibit their *de facto* employer from terminating a contract—and, hence, their employment—in response to efforts to unionize. By putting the benefits of collective bargaining beyond the reach of contract workers, the triangular nature of the employment relationship helps explain why contract employment, which otherwise resembles standard employment, is more

likely than standard work arrangements to be associated with bad job characteristics.

Policies that encompass the nonstandard employment relationship are needed to ensure accountability among those organizations that employ and benefit from nonstandard workers’ labor (Hiatt 1995). Laws protecting workers and legal definitions of employers and employees evolved simultaneously as a consequence of the labor movement’s fight against exploitation by employers. The observed growth of nonstandard employment relations underscores the need for policymakers to reconsider workplace-mediated social benefits (such as health insurance and pensions) and protections for nonstandard workers, including the right to organize.

Nonstandard work arrangements represent a potential source of employment flexibility for both employers and workers, and they are doubly attractive to employers because they often reduce employment costs. For many nonstandard workers, however, any gains in flexibility come at a high price, and for the society they are likely to exacerbate socioeconomic inequality if qualified workers who seek regular full-time jobs must settle for less desirable alternatives. Understanding the diversity of work arrangements and their consequences for job quality are increasingly important as we enter the twenty-first century.

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Appendix A. Weighted Means of Explanatory Variables, by Sex: Current Population Survey, February 1995

Explanatory Variable	Male	Female	Explanatory Variable	Male	Female
<i>Work Arrangement</i>			Associate degree	.079 †††	.097
[Regular full-time]	.751 †††	.665	BA degree	.174	.170
Standard	.694 †††	.620	Advanced degree	.096 †††	.078
Contingent	.037	.034	Attending school	.040 †	.046
Regular part-time	.076 †††	.215	Age	38.9	38.9
On-call and day labor	.013 †	.015		(12.3)	(12.1)
Temporary-help agency	.009 †††	.012	<i>City Type and Region</i>		
Contract company	.018	.009	[Suburban]	.406	.399
Independent contractor, wage and salary worker	.008 †††	.008	Urban	.219	.228
Independent contractor, self-employed	.065	.033	Rural	.218	.218
Other self-employed	.059 †††	.044	City type not identified	.157	.156
Employment insecurity	.076 ††	.084	[Northeast]	.187	.182
Unions/collective bargaining	.177 †††	.144	Midwest	.249	.242
Occupational complexity	.011 ††	.052	South	.343 ††	.370
	(.852)	(.777)	West	.221 †	.206
<i>Race/Ethnicity</i>			<i>Occupation</i>		
[White]	.787	.780	[Manager]	.149 †††	.134
Black	.092 †††	.120	Professional	.128 †††	.167
Hispanic	.093 †††	.073	Technical	.028 †††	.038
Other	.028	.028	Sales	.112	.114
Born outside the U.S.	.112 †††	.095	Clerical	.063 †††	.266
<i>Family Status</i>			Private household	.000 †††	.013
[Married, spouse employed]	.458 †††	.515	Protective services	.029 †††	.008
Married, spouse unemployed	.207 †††	.085	Other service occupations	.066 †††	.153
Single	.335 †††	.400	Craft	.190 †††	.024
Children	.411 †††	.430	Machine operatives	.072 †††	.052
<i>Education</i>			Transport operatives	.068 †††	.008
[High school grad]	.322 †††	.340	Laborers	.059 †††	.015
Less than high school grad	.121 †††	.082	Farm and forestry	.037 †††	.009
Some college	.209 †††	.234	<i>Industry</i>		
			[Manufacturing]	.222 †††	.118
			Agriculture, forestry, fishery	.035 †††	.013

(Continued on the next page)

(Appendix A continued from previous page)

Explanatory Variable	Male	Female	Explanatory Variable	Male	Female
Mining	.010 †††	.002	Business, repair services	.068 †††	.040
Construction	.093 †††	.013	Personal services	.018 †††	.031
Transportation	.095 †††	.050	Entertainment, recreational	.013	.013
Wholesale trade	.052 †††	.026	Professional services	.138 †††	.372
Retail trade	.151 †††	.173	Public administration	.055	.049
Finance, real estate	.050 †††	.085			
Private households	.001 †††	.016	Number of cases	12,747	10,434

Note: Numbers in parentheses are standard deviations (weighted).
†*p* < .05 ††*p* < .01 †††*p* < .001 (two tailed tests for sex differences)

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